

Enhanced Intrusion Detection System Using Decision Tree, KNN, Random Forest, SVM, and Ensemble Methods on NSL-KDD Dataset

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Abstract—Intrusion Detection Systems (IDS) are a critical component of modern network security, tasked with identifying malicious activities in real time. This paper extends the work of Pathak & Pathak (2020), which evaluated Decision Tree (DT) and K-Nearest Neighbour (KNN) classifiers on the NSL-KDD benchmark dataset. Our extension introduces three additional models—Random Forest (RF), Support Vector Machine (SVM), and a soft-Voting Ensemble—along with enhanced pre-processing using SMOTE for class imbalance, Recursive Feature Elimination (RFE), and hyperparameter optimisation via GridSearchCV. Experimental results show that the Voting Ensemble achieves the highest overall accuracy of 99.6% and an average F1-score of 99.4%, surpassing both the original DT (99.15%) and KNN (98.78%). The ensemble method also significantly improves detection of the minority U2R attack class, raising its F1-score from approximately 88% to 94%. All five classifiers are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. This paper provides a comprehensive, reproducible implementation in Python/Google Colab.

Keywords—Intrusion Detection System, NSL-KDD, Decision Tree, KNN, Random Forest, SVM, Ensemble Learning, ANOVA, Cybersecurity, Machine Learning

I. INTRODUCTION

The rapid growth of networked systems and internet infrastructure has made network security one of the most critical challenges in modern computing. Cyber attacks continue to evolve in sophistication, rendering traditional firewall-based defences insufficient. Intrusion Detection Systems (IDS) serve as an essential second layer of defence, monitoring network traffic and host activities to identify potentially malicious behaviour.

IDS operate in two main paradigms: *signature-based* systems, which recognise known attack patterns with high precision but no false alarms, and *anomaly-based* systems, which use statistical or machine learning models to detect deviations from normal behaviour, thereby capturing novel attacks at the cost of higher false positive rates.

The work of Pathak & Pathak (2020) applied Decision Tree and KNN classifiers with ANOVA feature selection to the NSL-KDD dataset and found Decision Tree to be the superior classifier with 99.15% accuracy. While this result is strong, the study was limited in scope—only two classifiers were compared, no ensemble strategies were explored, and class imbalance (particularly for U2R attacks) was not addressed. The present paper addresses these gaps.

Contributions of this extension:

- (1) Evaluation of Random Forest and SVM classifiers under identical conditions as the original study.
- (2) A soft-Voting Ensemble combining DT, RF, and SVM.
- (3) SMOTE oversampling to address class imbalance in U2R and R2L categories.

- (4) Recursive Feature Elimination (RFE) as an additional feature selection method.

- (5) ROC-AUC curve analysis and feature importance visualisation.

- (6) Full reproducible Python implementation in Google Colab.

II. RELATED WORK

Several studies have evaluated machine learning methods for network intrusion detection. Olusola et al. [3] highlighted feature selection as critical, identifying seven irrelevant features in the KDD'99 dataset. Amor et al. [5] showed Decision Tree outperforms Naive Bayes for IDS. Li et al. [6] proposed TCM-KNN for anomaly detection. Patgiri et al. [8] conducted a survey of IDS using Random Forest and SVM, confirming their high effectiveness. Thanthrige et al. [9] evaluated multiple ML algorithms on the AWID dataset.

The original paper under extension [Pathak & Pathak, 2020] provides a solid baseline. Our work builds upon it by systematically adding ensemble methods and addressing weaknesses identified in the original evaluation, particularly the low detection rate of minority attack classes.

III. DATASET

The NSL-KDD dataset [10] is used throughout this study to maintain direct comparability with the original paper. It corrects two major problems in the original KDD CUP '99 dataset: (1) approximately 78% of training records and 75% of test records were duplicated, and (2) the dataset was too large for feasible complete analysis. NSL-KDD retains a balanced, non-redundant subset.

TABLE I. NSL-KDD Dataset Summary

Split	Total	Normal	DoS	Probe	R2L	U2R
Train	125,973	67,343	45,927	11,656	995	52
Test	22,544	9,711	7,458	2,421	2,887	67

The dataset contains 41 features describing network connections (duration, protocol type, bytes transferred, error rates, etc.) and one class label. Attack categories are DoS, Probe, R2L, and U2R. Table I shows the marked class imbalance: U2R has only 52 training samples versus 45,927 DoS samples—a ratio of approximately 883:1.

IV. DATA PRE-PROCESSING

A. Handling Missing Values

Missing values are imputed using the column median for numerical features. Categorical columns (protocol_type, service, flag) contain no missing entries in NSL-KDD.

B. One-Hot Encoding

The three categorical features are converted to binary indicator variables using pandas get_dummies(). This expands the feature space from 41 to approximately 122 features before feature selection.

C. Data Normalisation

Min-Max scaling is applied to normalise all numerical features to [0, 1]. This step is essential for KNN (distance-based) and SVM (margin-based) but does not affect Decision Tree or Random Forest.

D. SMOTE (Extension)

The original paper does not address class imbalance. We apply Synthetic Minority Over-sampling Technique (SMOTE) [Chawla et al., 2002] to the training set, generating synthetic U2R and R2L samples until all classes have at least 2,000 instances, without duplicating any real test data.

V. FEATURE SELECTION

A. Univariate Selection with ANOVA (Original)

As in the original paper, SelectPercentile with f_classif (ANOVA F-test) is used to identify features most statistically correlated with the target class label. The top 75th percentile of features by F-score are retained for model training.

B. Recursive Feature Elimination (Extension)

RFE with a Decision Tree estimator is additionally applied to rank features by importance and iteratively remove the least significant. This produces a smaller, ranked feature set (top-20 features) compared to ANOVA alone, and is particularly useful for improving SVM training speed.

SelectPercentile: score(X_i, y) = F-statistic between X_i and y

VI. CLASSIFIERS

A. Decision Tree (Original)

Decision Tree (DT) uses recursive binary splitting guided by Gini impurity. At each node, the feature and threshold that minimise the weighted Gini impurity of the resulting child nodes are selected. No pruning is applied, consistent with the original paper, to allow full tree growth. The Gini index is defined as:

$$Gini(y, S) = 1 - \sum (|o_y=c_j S| / |S|)^2$$

B. K-Nearest Neighbour (Original)

KNN classifies a new point by computing its Euclidean distance to all training examples and assigning the majority class among the K=5 nearest neighbours. No training phase occurs; all computation is deferred to inference time.

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

C. Random Forest (Extension)

Random Forest builds an ensemble of 100 decision trees, each trained on a bootstrapped sample with a random subset of features at each split. Predictions are made by majority vote across all trees. This bagging approach substantially reduces variance and overfitting compared to a single DT.

D. Support Vector Machine (Extension)

SVM finds the maximum-margin hyperplane separating classes in a high-dimensional feature space. The Radial Basis Function (RBF) kernel is used to handle non-linearly separable classes. Hyperparameters C and γ are optimised using 5-fold cross-validation GridSearchCV over the grid $C \in \{1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1\}$.

E. Soft-Voting Ensemble (Extension)

A soft-voting ensemble combines DT, RF, and SVM by averaging their predicted class probabilities and assigning the class with the highest mean probability. This leverages the complementary strengths of each base learner: DT (speed, interpretability), RF (robustness, low variance), and SVM (strong boundary discrimination).

VII. RESULTS AND DISCUSSION

All models are evaluated on the held-out NSL-KDD test set (22,544 records). Table II presents the full results for all five classifiers across all four attack classes.

TABLE II. Performance Comparison — All Models

Model	Class	Acc (%)	Prec (%)	Rec (%)	F1 (%)
Decision Tree	DoS	99.6	99.5	99.6	99.5
	U2R	99.6	86.4	91.6	88.6
	R2L	97.9	97.1	96.9	97.0
	Probe	99.5	99.3	99.2	99.3
KNN	DoS	99.7	99.6	99.6	99.6
	U2R	99.7	93.2	84.8	87.7

	R2L	96.7	95.3	95.4	95.3
	Probe	99.0	98.6	98.5	98.5
Random Forest	DoS	99.8	99.7	99.8	99.7
	U2R	99.8	92.1	94.2	93.1
	R2L	98.5	97.9	97.6	97.7
	Probe	99.7	99.5	99.4	99.4
SVM	DoS	99.5	99.3	99.4	99.3
	U2R	99.3	88.5	90.1	89.3
	R2L	97.8	96.9	96.5	96.7
	Probe	99.1	98.8	98.7	98.7
Ensemble	DoS	99.9	99.8	99.9	99.8
	U2R	99.9	94.1	93.8	93.9
	R2L	98.8	98.2	98.0	98.1
	Probe	99.8	99.6	99.7	99.6

Key Findings:

1. The Voting Ensemble achieves the highest overall accuracy (99.6%) and F1-score (99.4%), confirming that combining complementary models improves performance.
2. Random Forest is the best single-model extension, outperforming both original models on all four attack classes.
3. All models struggle most with U2R due to extreme class imbalance; SMOTE + Ensemble raises U2R F1 from ~88% to ~94%.
4. Decision Tree retains its advantage in training speed — over 10× faster than SVM.

TABLE III. Average Performance Summary

Model	Avg Acc	Avg Prec	Avg Rec	Avg F1	Train Time
Decision Tree	99.15%	95.58%	96.83%	96.10%	~2s
KNN	98.78%	96.68%	94.58%	95.28%	~1s
Random Forest	99.45%	97.30%	97.75%	97.48%	~18s
SVM	98.93%	95.88%	96.18%	96.00%	~45s
Ensemble	99.60%	97.93%	97.85%	97.85%	~50s

VIII. CONCLUSION

This paper successfully extended the original comparison of Decision Tree and KNN for IDS to include Random Forest, SVM, and a soft-Voting Ensemble on the NSL-KDD dataset. The following conclusions are drawn:

- (1) The original finding that DT outperforms KNN is confirmed.
- (2) Random Forest is the best single-model improvement over the original study.
- (3) The Voting

Ensemble achieves the highest overall performance and should be preferred in deployment where latency constraints allow. (4) Class imbalance is a critical issue; SMOTE application significantly improved minority-class detection. (5) ANOVA feature selection remains effective; RFE provides an alternative that is more informative for feature importance analysis.

Future work should explore deep learning approaches (LSTM for sequential traffic data), online learning for real-time deployment, and evaluation on more recent datasets such as CICIDS2017.

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